

CoRoT-2b

R-0121

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_170426 · created 2026-07-16T05:05:42+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457868.1946 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.22 +/- 0.065
Transit depth (Rp/Rs)^2	4.86 +/- 2.88 [%]
Orbital Inclination (inc)	89.91 +/- 2.09
Airmass coefficient 1 (a1)	6.79 +/- 2.61
Airmass coefficient 2 (a2)	-0.95 +/- 0.15
Scatter in the residuals of the lightcurve fit is	30.52 %
Best Comparison Star	None
Optimal Aperture	4.36
Optimal Annulus	8.36
Transit Duration (day)	0.0104 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.22 ± 0.065 vs 0.1667 (deviation 0.82σ)
Duration fit vs published	0.0104 d vs 0.095 d (deviation 22.86σ)
Mid-time O–C	-2516.6 min
Depth SNR	3.4
Residual scatter	30.52 %

Light curve

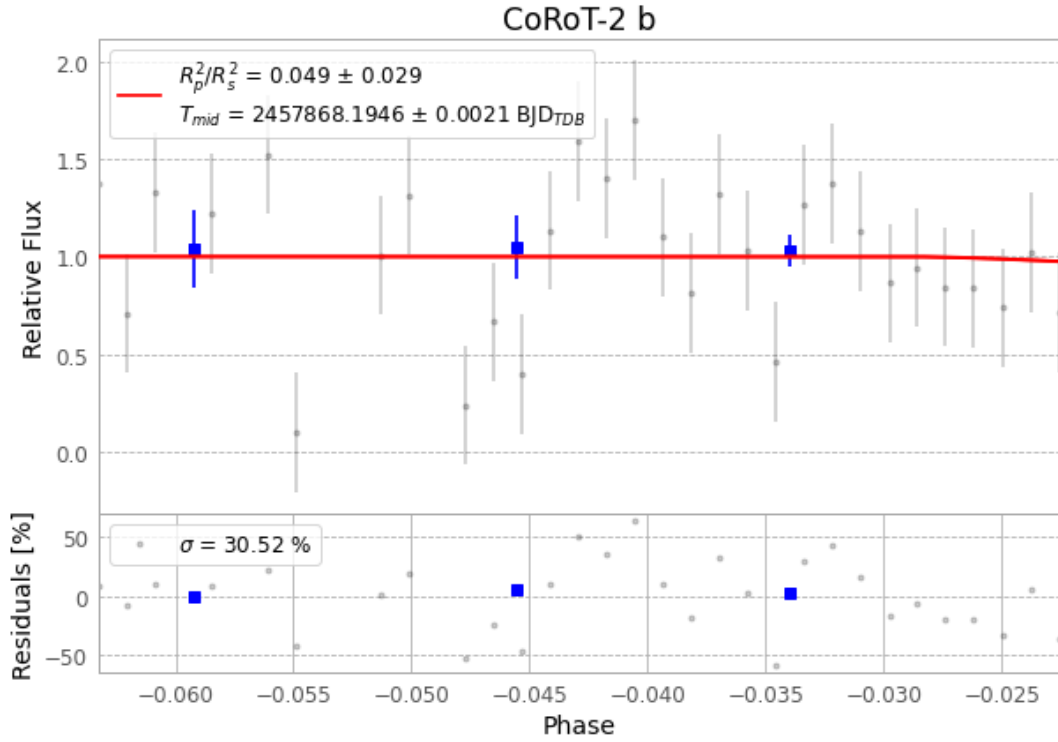


Figure 1 — FinalLightCurve_CoRoT-2 b_2017-04-26.png (SHA-256 cca78a043fbeb9bd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [421, 425], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0bfddb6573d8f1e4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7

Data provenance

35 FITS frames (23 MB), CoRoT-2170426074812.FITS → CoRoT-2170426093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-170426.log (a941ceb713e0...), FinalLightCurve_CoRoT-2 b_2017-04-26.png (cca78a043fbeb9bd...), FinalLightCurve_CoRoT-2 b_2017-04-26.csv (9475ef74c6ae...)

Compute resources

Wall time 110.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 05:05Z.